

Data Science Report

A Comparative Study of Classifiers and Preprocessing Strategies on Parkinson's Disease and Forest Covertype Datasets

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June 7, 2026

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1 Introduction

The purpose of this project is to evaluate the impact of different preprocessing strategies and classification algorithms for two classification problems: predicting Parkinson’s disease (PD) from speech recordings and predicting forest cover type (CT) from cartographic variables.

To investigate these problems, two datasets with markedly different characteristics are analysed. The Parkinson’s disease dataset represents a high-dimensional, low-sample-size problem, whereas the forest cover type dataset is characterised by a relatively low number of features and a large number of observations.

Several preprocessing techniques will be explored, including feature scaling, feature selection, dimensionality reduction, and class balancing. Six classification algorithms will be evaluated: Naive Bayes, K-Nearest Neighbors (KNN), Decision Trees, Random Forests, Gradient Boosting, and XGBoost. Model performance will be assessed using multiple evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC.

The following sections describe the datasets, define the experimental methodology, present the results, and discuss the main findings.

2 Datasets

2.1 Parkinson’s Disease Dataset

The Parkinson’s disease dataset contains 756 voice recordings collected from 252 individuals, with three recordings available for each participant. Each record contains 754 features, including a patient identifier, a binary gender variable, and 752 acoustic features extracted from the recordings.

Among the 252 participants, 64 are healthy controls and 188 have Parkinson’s disease. Consequently, 564 recordings (74.6%) correspond to individuals with PD, while 192 recordings (25.4%) belong to healthy controls, resulting in a class imbalance ratio of approximately 3:1.

No missing values are present in the dataset; therefore, data imputation is not required.

This dataset has two important characteristics that must be considered in the analysis. First, the number of features is almost equal to the number of observations. This increases the risk of overfitting and makes dimensionality an important challenge. Second, each participant contributes multiple recordings, meaning that the dataset contains repeated measurements from the same individuals. This structure must be considered carefully when evaluating model performance.

2.2 Covertypes Dataset

The Covertypes dataset contains 581,012 records and 54 features. Each record describes a 30-by-30-metre forest patch using cartographic variables derived from twelve original attributes encoded across 54 columns.

Ten of the original attributes are continuous physical measurements, including elevation, slope, aspect, three hillshade indices measured at different times of day, and horizontal or vertical distances to nearby hydrology, roadways, and fire-ignition points. The remaining two attributes are categorical: wilderness area, with four possible levels, and soil type, with forty possible levels. These categorical variables are already one-hot encoded into binary columns.

All feature values are stored as integers. The continuous measurements are represented in their natural integer units, such as metres, degrees, and 0–255 hillshade values, while the one-hot encoded columns take values of either 0 or 1.

The target variable is the forest cover type, which represents the dominant tree species present in each forest patch. This is a multiclass classification problem with seven possible classes: Spruce/Fir, Lodgepole Pine, Ponderosa Pine, Cottonwood/Willow, Aspen, Douglas-fir,

and Krummholz. The original dataset is highly imbalanced. Lodgepole Pine is the most frequent class, with 283,301 records, corresponding to 48.8% of the dataset, followed by Spruce/Fir, with 211,840 records, corresponding to 36.5%. In contrast, some cover types are much less represented, such as Cottonwood/Willow, which accounts for only 2,747 records, or 0.47% of the dataset.

For the experiments, a balanced subsample of 7,000 records is used, containing 1,000 observations from each cover type. As a result, each class represents 14.3% of the subsampled dataset. This subsample defines the working version of the Covertypes dataset used in the modelling experiments.

No missing values are present in the Covertypes dataset; therefore, data imputation is not required. Since the categorical variables are already represented using one-hot encoded columns, no additional categorical encoding is necessary.

Table 1: Class distribution in the raw Covertypes dataset.

Cover type	Records	Percentage
Spruce/Fir	211,840	36.5%
Lodgepole Pine	283,301	48.8%
Ponderosa Pine	35,754	6.2%
Cottonwood/Willow	2,747	0.5%
Aspen	9,493	1.6%
Douglas-fir	17,367	3.0%
Krummholz	20,510	3.5%
Total	581,012	100%

The balanced subsample is used to reduce the effect of the strong class imbalance in the original dataset and to make the experiments more computationally manageable. However, this choice also means that many observations from the majority classes are discarded, which may reduce how representative the subsample is of the original dataset. Scaling will be considered for the continuous cartographic variables, especially because distance-based models such as K-Nearest Neighbors are sensitive to feature magnitudes. The one-hot encoded variables will be kept as binary indicators.

3 Methodology

3.1 Experimental Overview

The methodology is designed to compare classification algorithms and preprocessing strategies across two datasets with different characteristics. The objective is not only to identify the best-performing model, but also to understand how different preprocessing techniques affect model performance under different data conditions.

3.2 Dataset Preparation

Before training the models, the input features and target variable will be separated for each dataset.

For the Parkinson’s disease dataset, the patient identifier will be removed from the predictive features because it does not represent a meaningful clinical or acoustic predictor and could lead to data leakage. However, the patient identifier will be retained separately for patient-level splitting and grouped cross-validation.

For the Covertypes dataset, the original class distribution is highly imbalanced. To reduce the effect of this imbalance and make the experiments more computationally manageable, a balanced

subsample of 7,000 records will be used. This subsample will contain 1,000 observations from each of the seven cover types, meaning that each class will represent 14.3% of the working dataset. Therefore, the reported Covertypes results should be interpreted as performance on a balanced version of the dataset rather than on the original naturally imbalanced distribution.

3.3 Preprocessing Strategies

Several preprocessing strategies will be investigated to evaluate their impact on classifier performance. These include feature scaling, feature selection, dimensionality reduction, and class balancing.

Feature scaling will be applied because several variables are measured on different scales. This is especially important for distance-based classifiers such as K-Nearest Neighbors, where features with larger numerical ranges can dominate the distance calculation. Scaling is also required before applying PCA, since PCA is sensitive to the scale of the input variables. Although tree-based models are generally less sensitive to feature scaling, scaled versions of the data will still be evaluated across all classifiers to compare the effect of scaling between model families.

Feature selection will be explored to reduce the number of predictors by retaining the most informative features. This is particularly relevant for the Parkinson’s disease dataset, where the number of features is almost equal to the number of observations. Reducing the feature space may help limit overfitting, improve generalization, and make the models easier to interpret.

Principal Component Analysis (PCA) will be evaluated as a dimensionality reduction technique. Unlike feature selection, PCA transforms the original features into a smaller set of uncorrelated components. This may reduce redundancy and noise in high-dimensional data, although it also reduces interpretability because the resulting components are combinations of the original variables.

Class balancing is not applied as a separate preprocessing step in this study. For the Parkinson’s disease dataset, the class imbalance is mild (approximately 3 to 1) and is left unaddressed at the preprocessing stage; its effect on per-class metrics is examined directly in the evaluation. For the Forest Covertypes dataset, the experiments are conducted on the balanced subsample described in Section 2.2, so no additional class balancing is required.

No imputation will be required for either dataset, since no missing values are present. In addition, no further categorical encoding is required for the Covertypes dataset because the wilderness area and soil type variables are already represented using one-hot encoded columns.

3.4 Model Development Pipeline

For each dataset, an 80/20 train-test split will be used. The test set will be kept separate and used only for the final evaluation of the selected model. All model comparison, preprocessing comparison, and hyperparameter tuning will be performed using cross-validation on the training set.

For the Parkinson’s disease dataset, the split will be performed at the patient level, ensuring that all recordings from the same individual are assigned either to the training set or to the test set. Grouped cross-validation will then be used on the training set so that recordings from the same patient do not appear in both training and validation folds.

For the Covertypes dataset, a stratified 80/20 train-test split will be used to preserve the balanced class distribution across training and test sets. Stratified cross-validation will then be used on the training set during model comparison and hyperparameter tuning.

3.5 Experimental Configurations

This section defines the baseline models, preprocessing configurations, classification algorithms, and experimental procedure used to compare model performance.

3.5.1 Baseline Models

Baseline models will first be used to provide a simple reference point for evaluating classifier performance. For the Parkinson’s disease dataset, the baseline model will predict the majority class. Since approximately 74.6% of recordings correspond to individuals with Parkinson’s disease, this baseline is expected to achieve an accuracy close to this value, despite having limited practical value.

For the Covertypes dataset, the balanced working dataset contains 1,000 observations per class. Therefore, a simple majority-class baseline would achieve approximately 14.3% accuracy. This baseline provides a lower reference point against which the trained multiclass classifiers can be compared.

3.5.2 Preprocessing Configurations

Several preprocessing configurations will be evaluated for each dataset.

For the Parkinson’s disease dataset, the configurations will include raw features, scaled features, scaled features with feature selection, and scaled features with PCA. This allows the effects of scaling, dimensionality reduction, and feature selection to be evaluated separately and in combination.

For the Covertypes dataset, the configurations will include raw features, scaled continuous features, and scaled features with feature selection. Since the dataset contains both continuous cartographic variables and one-hot encoded categorical variables, scaling will mainly be applied to the continuous variables. PCA will not be used as a main preprocessing strategy for this dataset because many of the features are one-hot encoded categorical variables, making the resulting components harder to interpret.

3.5.3 Classification Algorithms

Each preprocessing configuration will be evaluated using the same set of classification algorithms: Naive Bayes, K-Nearest Neighbors, Decision Trees, Random Forests, Gradient Boosting, and XGBoost. Using the same classifiers across preprocessing configurations allows the effect of preprocessing to be compared consistently across different model families.

3.5.4 Two-Stage Experimental Procedure

The experiments will be performed in two stages. In the first stage, all preprocessing and classifier combinations will be evaluated using fixed/default classifier settings through cross-validation on the training set. This stage is used to compare the effect of preprocessing strategies without using the held-out test set.

In the second stage, hyperparameter tuning will be performed only for the most promising preprocessing-classifier combinations identified in the first stage. Grid search will be used for this tuning process, again using cross-validation on the training set. The final selected model for each dataset will then be evaluated once on the held-out test set.

3.6 Model Evaluation

For the Parkinson’s disease dataset, performance will be evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices. Since the dataset is imbalanced, accuracy alone may be misleading. Recall and F1-score are especially important because they help assess whether the model correctly identifies individuals with Parkinson’s disease rather than simply favouring the majority class.

For the Covertypes dataset, accuracy will be reported together with macro-averaged F1-score and per-class precision, recall, and F1-score. Since this is a multiclass problem, these metrics

help identify whether a model performs consistently across all forest cover types rather than only achieving high overall accuracy.

4 Results

4.1 Baseline Results

A majority-class baseline was evaluated for each dataset to provide a simple reference point for the trained classifiers. For the Parkinson’s disease dataset, predicting the majority class (PD) on every recording yields an accuracy of 0.746, matching the class proportion. For the Covertypes dataset, where the balanced subsample contains 1,000 records of each of the seven cover types, any single-class prediction yields an accuracy of approximately 0.143.

Table 2: Majority-class baseline accuracy for each dataset.

Dataset	Baseline strategy	Accuracy
Parkinson’s	Predict PD for every recording	0.746
Covertypes	Predict any single cover type	0.143

4.2 Parkinson’s Disease Results

This section focuses on the Parkinson-specific questions: whether scaling helped, whether feature selection or PCA reduced overfitting, which model performed best overall, and how the imbalanced class structure affected the per-class metrics.

4.2.1 Overall Model Comparison

Table 3: Overall comparison of the six classifiers on Parkinson’s, each evaluated with its Stage-2 best preprocessing and tuned hyperparameters. Metrics are 10-fold patient-grouped cross-validation means on the training set.

Model	Best preprocessing	Accuracy	Precision	Recall	F1	ROC-AUC
Naive Bayes	scaled + FS	0.7702	0.8628	0.8290	0.8424	0.7783
kNN	scaled + PCA	0.8146	0.8297	0.9539	0.8859	0.7601
Decision Tree	raw	0.7627	0.8297	0.8658	0.8439	0.6629
Random Forest	raw	0.8295	0.8401	0.9607	0.8949	0.8229
Gradient Boosting	scaled	0.8379	0.8465	0.9629	0.8997	0.8224
XGBoost	scaled + PCA	0.8196	0.8393	0.9476	0.8880	0.7999

The four top classifiers by F1 — Gradient Boosting (0.900), Random Forest (0.895), XGBoost (0.888), and kNN (0.886) — cluster within 0.014 of each other, while Decision Tree (0.844) and Gaussian Naive Bayes (0.842) trail by 5–6 points. All six clear the majority-class baseline of $F1 = 0.854$ established in §4.1, but only by ≤ 0.05 for the four that do — a gap that is real but modest. The ROC-AUC column reorders the leaderboard substantially: Random Forest and Gradient Boosting are tied at the top (0.823 and 0.822), but kNN — despite a competitive F1 — drops to last place among the ensemble-level classifiers at 0.760, behind Naive Bayes. This is a property of the tuned kNN configuration ($k = 1$, Chebyshev): with a single neighbor the classifier’s `predict_proba` outputs are essentially $\{0, 1\}$, leaving ROC-AUC’s threshold sweep no score gradient to operate on. Decision Tree exhibits the same effect more severely (ROC-AUC = 0.663, the lowest in the table). Finally, the Recall column is uniformly high (≥ 0.86 across all six classifiers, ≥ 0.95 across the four leaders) while the implicit specificity

(healthy-class recall) is much lower: every classifier predicts PD aggressively, a direct consequence of the dataset’s 75/25 class imbalance that is examined in detail in §4.2.5 and §4.2.6.

4.2.2 Effect of Scaling

Table 4: Effect of scaling on Parkinson’s: cross-validation mean F1 for each classifier under raw and scaled features.

Model	Raw F1	Scaled F1	Difference
Naive Bayes	0.7702	0.7913	+0.0210
kNN	0.8185	0.8774	+0.0589
Decision Tree	0.8321	0.8202	−0.0119
Random Forest	0.8949	0.8868	−0.0081
Gradient Boosting	0.8855	0.8895	+0.0040
XGBoost	0.8833	0.8803	−0.0029

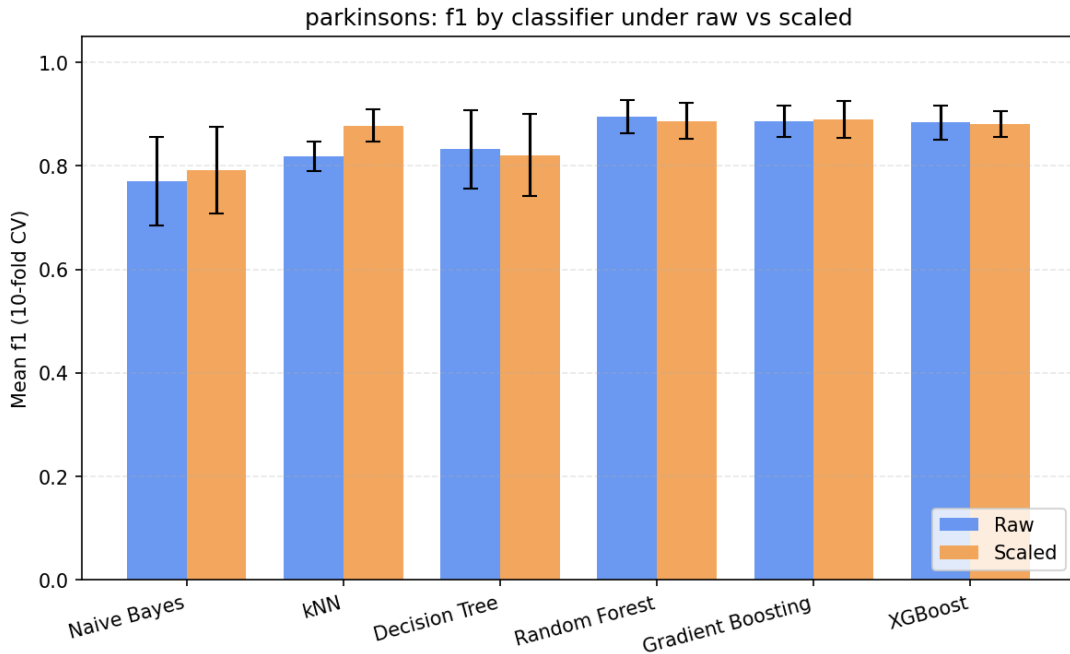


Figure 1: F1-score for each classifier on Parkinson’s under raw versus scaled preprocessing (10-fold patient-grouped cross-validation, default hyperparameters).

Feature scaling produces a substantial benefit only for the distance-based classifier: kNN improves by approximately six F1 points (0.818 to 0.877). Naive Bayes shows a modest improvement of about two points (0.770 to 0.791). The four tree-based methods are essentially unaffected (differences within ± 0.012 F1), as expected — tree splits compare features against thresholds and are mathematically scale-invariant. The kNN improvement is the largest single preprocessing effect observed in either dataset, reinforcing the principle that scaling is essential for distance-based methods when features span different numerical ranges.

4.2.3 Feature Selection Analysis

Feature selection was investigated using SelectKBest with ANOVA F-statistic scoring. The number of selected features k was swept across 18 values from 1 to 750 to capture the entire range from a single feature to the full feature set. All six classifiers were evaluated at each value of

k using ten-fold patient-grouped cross-validation, with five metrics tracked: accuracy, precision, recall, F1-score, and ROC-AUC, along with specificity (recall on the healthy control class).

Table 5: Feature selection at $k = 50$ versus the scaled baseline. At a single fixed k , feature selection produces a notable improvement only for Naive Bayes; other classifiers either remain unchanged or degrade slightly.

Model	Scaled F1	Scaled + FS F1	Difference
Naive Bayes	0.7913	0.8424	+0.0511
kNN	0.8774	0.8514	−0.0260
Decision Tree	0.8202	0.8158	−0.0044
Random Forest	0.8868	0.8820	−0.0048
Gradient Boosting	0.8895	0.8795	−0.0100
XGBoost	0.8803	0.8756	−0.0048

At a single chosen value of $k = 50$, feature selection improves Naive Bayes by approximately five F1 points ($0.791 \rightarrow 0.842$) but provides no benefit and minor harm to the other five classifiers. This pattern, taken in isolation, suggests feature selection is largely ineffective for the Parkinson’s dataset. However, examining the full sweep across all values of k reveals a richer picture.

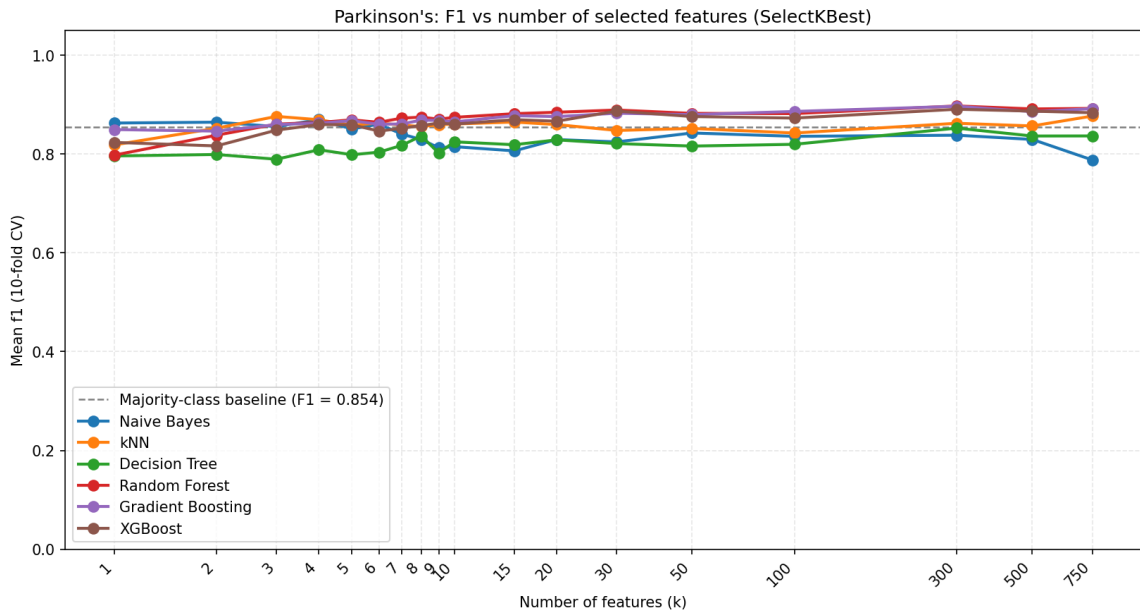


Figure 2: F1-score versus number of selected features for all six classifiers on Parkinson’s. The dashed line marks the majority-class baseline of $F1 = 0.854$ — the F1 a predictor that always outputs Parkinson’s would achieve due to the 75/25 class imbalance.

Figure 2 plots F1 as a function of the number of selected features. The curves are nearly flat across the entire k range: F1 varies by less than 0.05 points between $k = 1$ and $k = 750$ for every classifier. Several classifiers (Random Forest, Gradient Boosting, XGBoost) sit slightly above the majority-class baseline of 0.854 across the full range. Decision Tree and Naive Bayes fluctuate around it. Taken at face value, this would suggest that feature selection has minimal effect.

However, this view is misleading because the F1 metric is dominated by the class imbalance: a trivial predictor that always outputs Parkinson’s would already achieve $F1 = 0.854$. The fact that most classifiers are within 0.04 of this baseline at most values of k indicates not that the classifiers are skilled, but that the baseline itself is high. To see whether the classifiers are actually learning, a metric immune to class imbalance is required.

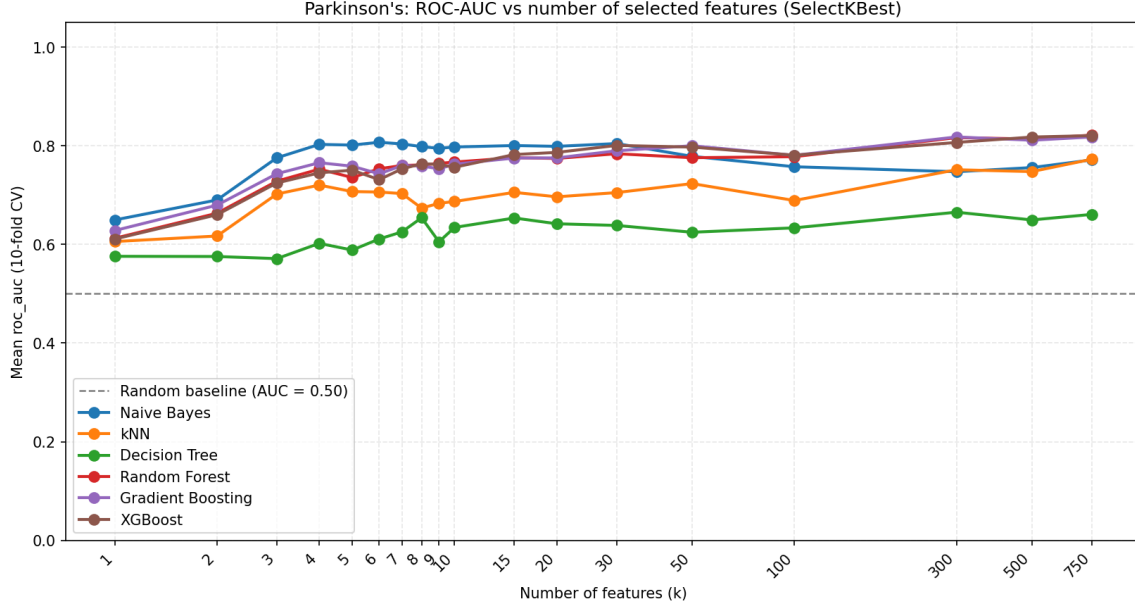


Figure 3: ROC-AUC versus number of selected features. ROC-AUC is invariant to class imbalance — a random classifier achieves 0.50 regardless of class proportions. The dashed line marks this random baseline.

Figure 3 plots ROC-AUC against k . Unlike F1, ROC-AUC is invariant to class imbalance: a random predictor scores 0.50 regardless of class proportions. The curves now reveal clear structure. With a single feature, all classifiers score between 0.58 and 0.65 — modestly better than random, but signalling weak discrimination. From $k = 1$ to $k = 10$, ensemble methods rise steeply to approximately 0.78. Beyond $k = 15$ the curves plateau, with marginal gains as k grows further. Naive Bayes peaks early (around $k = 5$ at ROC-AUC 0.80) and stays near that level. Decision Tree alone fails to climb substantially, remaining around 0.60 ROC-AUC throughout, consistent with its known sensitivity to high-dimensional noise.

This figure reveals that the signal in the Parkinson’s dataset is concentrated in the first ten or so features identified by ANOVA F-scores. Going beyond $k = 15$ captures almost no additional discriminative information for most classifiers.

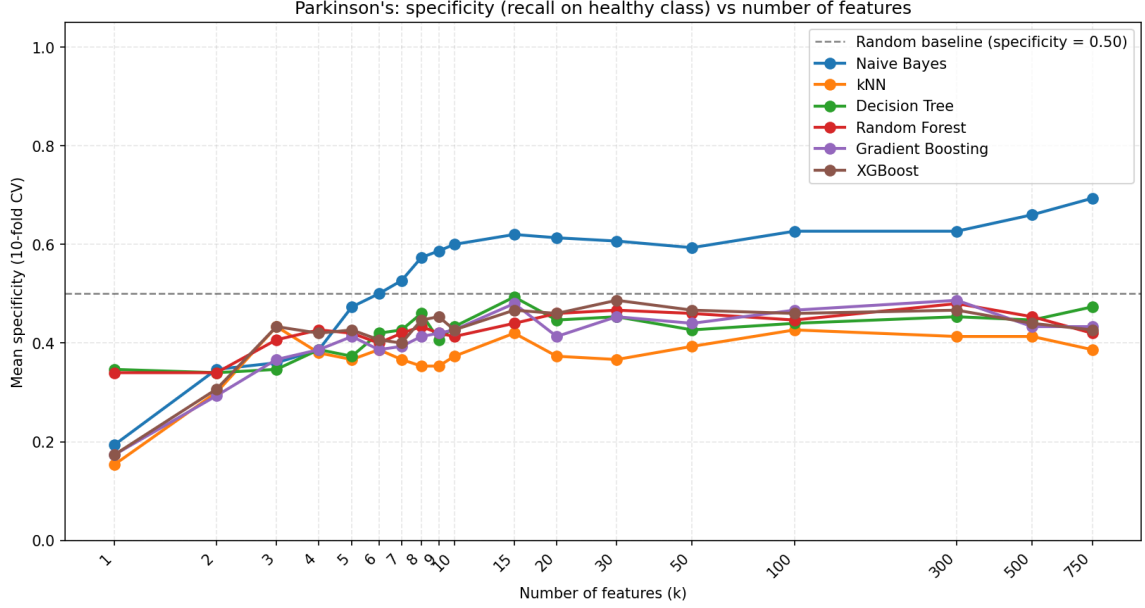


Figure 4: Specificity (recall on the healthy control class) versus number of selected features. The dashed line marks the random baseline of 0.50.

Figure 4 plots specificity — the fraction of actual healthy controls correctly identified as healthy. The pattern reveals a systematic problem masked by the F1 view: only Naive Bayes consistently exceeds the random-baseline specificity of 0.50 (reaching approximately 0.69 at $k = 750$). All tree-based ensembles plateau below 0.50, in the band 0.40–0.48, regardless of how many features are selected. This means that across the full feature-count range, the ensemble classifiers misclassify more than half of healthy controls as having Parkinson’s. The high sensitivity values that drive their high F1 scores are achieved by adopting a strong bias toward predicting Parkinson’s — they catch nearly all positive cases by predicting the positive class often.

Feature selection alone does not break this class-bias regime. Even at $k = 750$, when all features are available, specificity for tree-based ensembles remains in the 0.40–0.48 range. The bias is structural — driven by the 3:1 class imbalance in training data — and cannot be addressed by feature engineering.

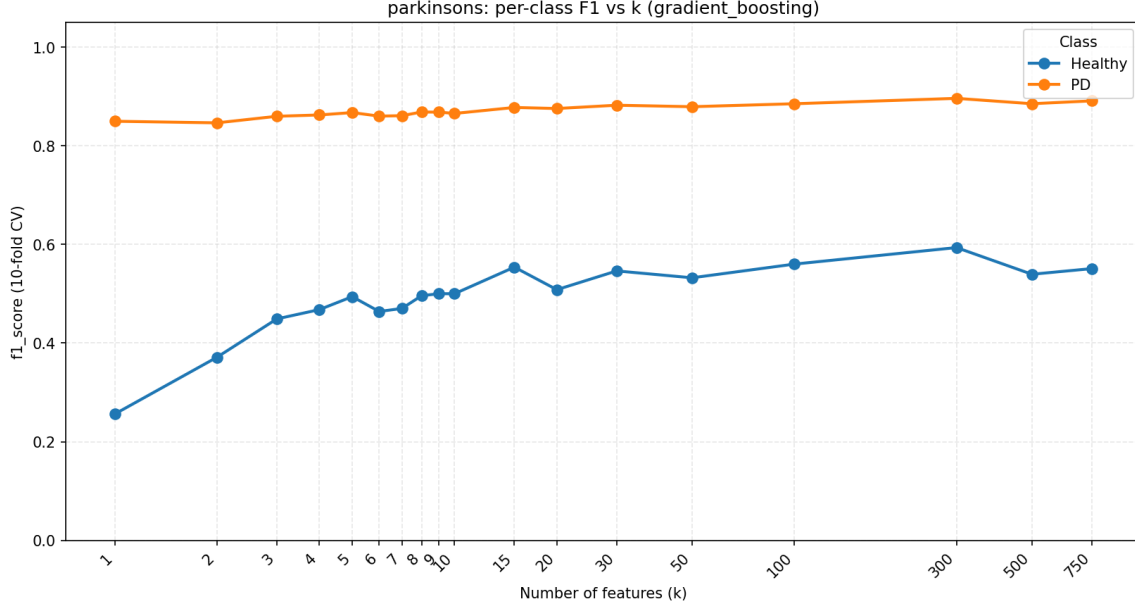


Figure 5: Per-class F1 for Gradient Boosting (the overall best Parkinson’s classifier) as a function of the number of selected features. The two classes are shown as separate lines: Parkinson’s recordings (orange) and healthy controls (blue).

Figure 5 decomposes the Gradient Boosting classifier’s F1 by class. The Parkinson’s-class F1 sits between 0.85 and 0.89 across the entire k range — high because the model correctly identifies most actual Parkinson’s recordings. The healthy-class F1 climbs more steeply from 0.26 at $k = 1$ to approximately 0.55 by $k = 15$, then plateaus and never exceeds 0.60. The persistent gap of approximately 0.30 F1 between the two lines visualises the class-bias finding numerically: feature selection narrows the gap somewhat at low k but does not eliminate it.

Which features drive the signal? To understand why the discriminative signal saturates so quickly, Table 6 lists the 20 most discriminative features ranked by their ANOVA F-statistic on the training set. The pattern is striking: 95% of the top 20 features come from just two families — MFCC delta coefficients, which measure how the spectral envelope of speech changes between consecutive frames, and TQWT (Tunable-Q Wavelet Transform) decomposition statistics, which capture multi-resolution time-frequency properties at specific decomposition levels. Beyond $k \approx 15$, additional features are predominantly correlated variants of measurements that the top-ranked features already capture — different MFCC coefficients of the same speech segments, or TQWT statistics at neighbouring decomposition levels. This concentration explains the ROC-AUC plateau quantitatively.

Table 6: The 20 most discriminative individual features in the Parkinson’s training set, ranked by ANOVA F-statistic.

Rank	Feature	Family	F-score
1	mean_MFCC_2nd_coef	MFCC cepstral	118.1
2	tqwt_kurtosisValue_dec_27	TQWT	112.8
3	tqwt_kurtosisValue_dec_26	TQWT	99.5
4	std_9th_delta_delta	MFCC delta	99.4
5	std_8th_delta_delta	MFCC delta	98.3
6	std_7th_delta_delta	MFCC delta	93.1
7	std_6th_delta_delta	MFCC delta	92.0
8	tqwt_stdValue_dec_12	TQWT	87.7
9	std_8th_delta	MFCC delta	86.8
10	std_9th_delta	MFCC delta	85.5
11	tqwt_minValue_dec_12	TQWT	84.8
12	tqwt_maxValue_dec_12	TQWT	82.2
13	std_6th_delta	MFCC delta	81.0
14	tqwt_entropy_log_dec_12	TQWT	77.4
15	tqwt_stdValue_dec_11	TQWT	77.4
16	std_11th_delta_delta	MFCC delta	77.2
17	std_7th_delta	MFCC delta	76.0
18	tqwt_entropy_shannon_dec_16	TQWT	74.7
19	std_10th_delta_delta	MFCC delta	73.5
20	tqwt_entropy_shannon_dec_15	TQWT	73.0

Three findings summarise the feature-selection analysis for Parkinson’s. First, the discriminative signal is concentrated in approximately the first ten features ranked by ANOVA F-statistic: ROC-AUC plateaus by $k = 15$ for ensemble methods, a plateau the top-features table above explains mechanistically. Second, the F1 metric is misleading in this setting because of class imbalance — a trivial baseline already scores 0.854, leaving little headroom for visible classifier improvement. Third, feature selection does not address the underlying class-bias problem — most classifiers continue to misidentify healthy controls regardless of k .

4.2.4 PCA Analysis

Principal Component Analysis was investigated as an alternative dimensionality reduction approach. The number of principal components was swept across eight values from 5 to 500, with all six classifiers evaluated at each value using ten-fold patient-grouped cross-validation.

Table 7: PCA at 95% variance (approximately 158 components) versus the scaled baseline.

Model	Scaled F1	Scaled + PCA F1	Difference
Naive Bayes	0.7913	0.8271	+0.0358
kNN	0.8774	0.8786	+0.0012
Decision Tree	0.8202	0.8098	−0.0104
Random Forest	0.8868	0.8641	−0.0227
Gradient Boosting	0.8895	0.8770	−0.0125
XGBoost	0.8803	0.8903	+0.0100

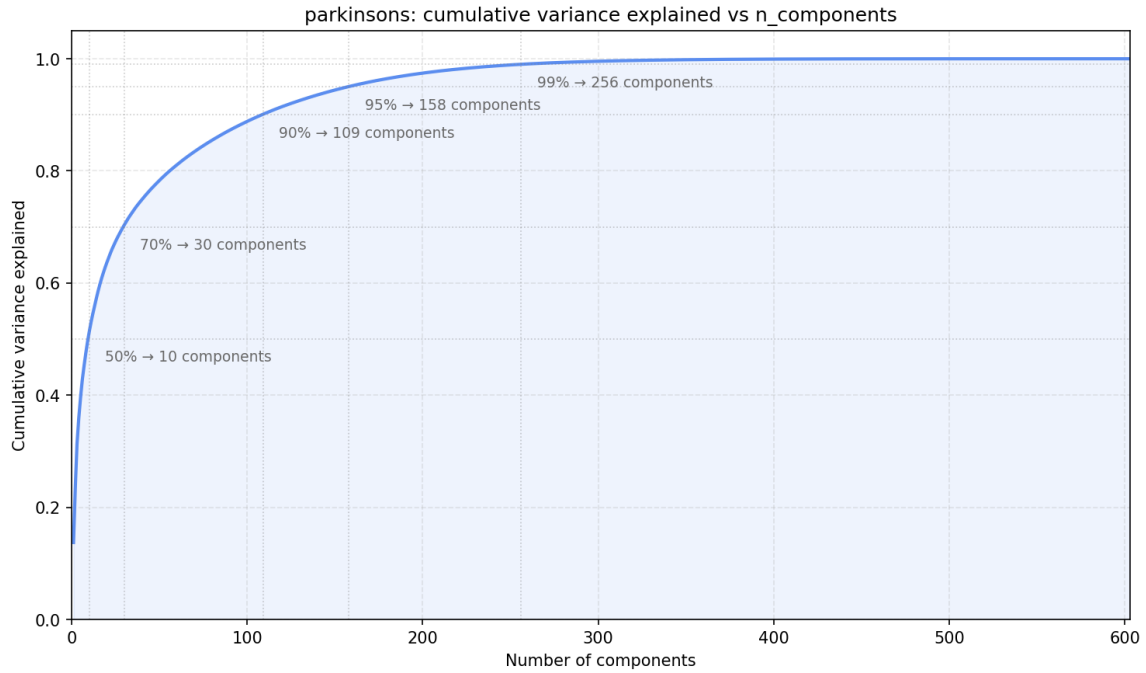


Figure 6: Cumulative variance explained as a function of the number of principal components for the Parkinson’s training data. The plot annotates the number of components needed to reach 50%, 70%, 90%, 95%, and 99% of total variance.

Figure 6 confirms that the Parkinson’s dataset is highly redundant: 50% of total variance is captured in just 10 components, 90% in 109 components, and 99% in 256 components. This is consistent with the structure of the voice-acoustic features: many measurements (jitter variants, shimmer variants, MFCC coefficients at different frames) derive from the same underlying signal and are therefore highly correlated.

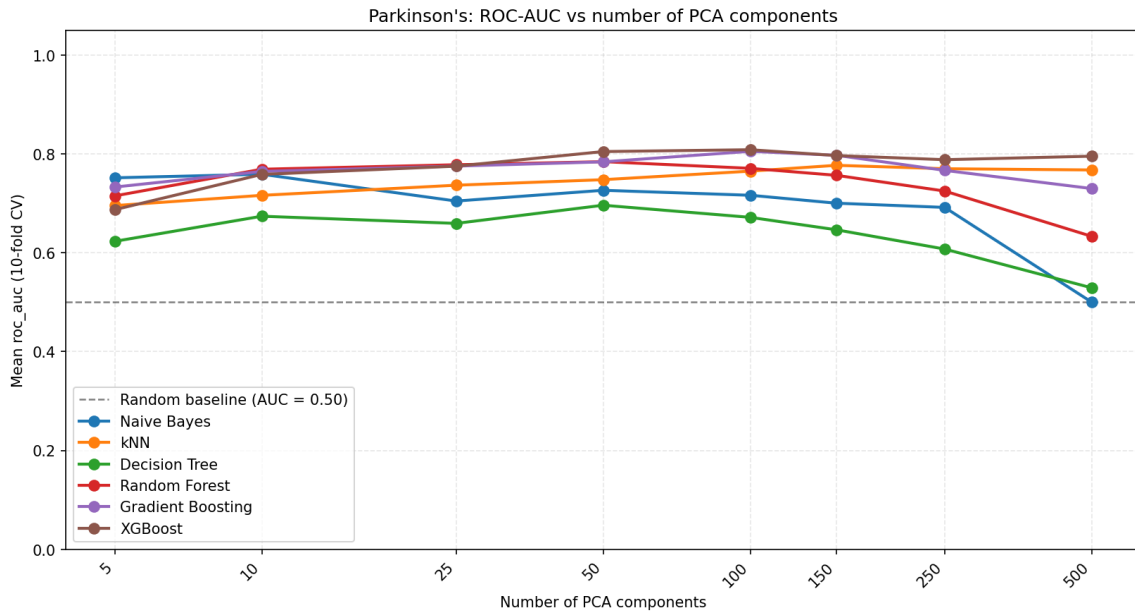


Figure 7: ROC-AUC versus number of PCA components, for all six classifiers on Parkinson’s. The dashed line marks the random-baseline ROC-AUC of 0.50.

Figure 7 reveals a striking failure mode at the high end of the component range. From 5 to approximately 250 components, ROC-AUC for all classifiers stays in the 0.70–0.80 band. At

500 components — beyond the 256-component threshold where 99% of variance is captured — the remaining “components” are essentially noise. Naive Bayes collapses to ROC-AUC of 0.50 (literally random discrimination), Decision Tree falls to 0.53, and Random Forest drops from 0.78 to 0.63. kNN and XGBoost remain relatively stable.

This is a cautionary finding: PCA is not a free transformation, and applying it with too many components is actively harmful for several model families, particularly methods that assume feature independence (Naive Bayes) or that overfit to weak signals (single-tree models). A sensible operating range is 50–150 components, corresponding to approximately 80–92% of variance retained.

What do the top principal components represent? The feature-ranking analysis of Table 6 ordered individual features by their univariate class-separation power (an ANOVA F-statistic) and showed that MFCC delta and TQWT features carry the strongest discriminative signal. PCA orders *directions* by variance, which is a different criterion. Table 8 reports, for each of the first ten principal components, the single feature with the largest absolute loading on that component and the feature family that dominates its top-10 loadings. The pattern is informative: the first six PCs — accounting for roughly 43% of total variance — are entirely Wavelet- and TQWT-dominated; MFCC delta does not become a major contributor to any component until PC7, by which point still only 45% of variance has been captured. The largest modes of variation in the scaled feature space describe wavelet- and TQWT-domain structure shared across recordings, while the MFCC delta features that drive the strongest univariate separation contribute comparatively little to overall variance. This mismatch between “variance-dominant” and “discrimination-dominant” directions helps explain why PCA only meaningfully improves Naive Bayes in Table 7: projecting onto top components preserves variance but does not preferentially preserve the directions that separate classes.

Table 8: Top contributing feature and dominant feature family of the first ten principal components of the (scaled) Parkinson’s training set. The dominant family is the most frequent family among the ten features with the largest absolute loading on each component.

PC	Var. (%)	Top feature (largest loading)	Dominant family (top-10)
1	13.73	app_LT_entropy_shannon_5_coef	Wavelet
2	9.56	tqwt_maxValue_dec_8	TQWT
3	8.02	det_entropy_log_2_coef	Wavelet
4	4.56	tqwt_minValue_dec_31	TQWT
5	3.66	tqwt_entropy_shannon_dec_5	TQWT
6	3.17	tqwt_maxValue_dec_21	TQWT
7	2.48	tqwt_energy_dec_6	MFCC delta
8	2.29	tqwt_stdValue_dec_28	TQWT
9	2.07	det_LT_entropy_shannon_5_coef	TQWT
10	1.82	tqwt_kurtosisValue_dec_4	TQWT

4.2.5 Hyperparameter Sweep Analysis

For each tunable classifier (kNN, Decision Tree, Random Forest, Gradient Boosting, and XGBoost) the two principal hyperparameters were swept using 10-fold patient-grouped CV, with the secondary hyperparameter shown as separate lines on each plot and the remaining hyperparameters held at their defaults. Each classifier was evaluated on its Stage-1 best preprocessing configuration. Naive Bayes is omitted: the variant used (Gaussian Naive Bayes) has no tunable hyperparameters in the search space considered. For every sweep cell, all six metrics — accuracy, precision, recall, specificity, F1, and ROC-AUC — are recorded; reporting them side-by-side

rather than F1 alone is important here because, as the following plots show, the choice of metric noticeably changes which configuration appears optimal.

k-Nearest Neighbors. Figure 8 sweeps the number of neighbors k across 30 values from 1 to 150, with one line per distance metric. The distance metric is not just a tuning detail: Manhattan distance peaks at very small k ($k = 3$) and degrades steadily as k grows, while Chebyshev distance behaves in the opposite direction, peaking around $k = 35$ –80. Euclidean sits in between. Specificity collapses toward zero at small k for all three metrics — a sign that the closest neighbors are overwhelmingly PD patients in this 75/25-imbalanced dataset. Best F1: $k = 35$, Chebyshev, $F1 = 0.886$.

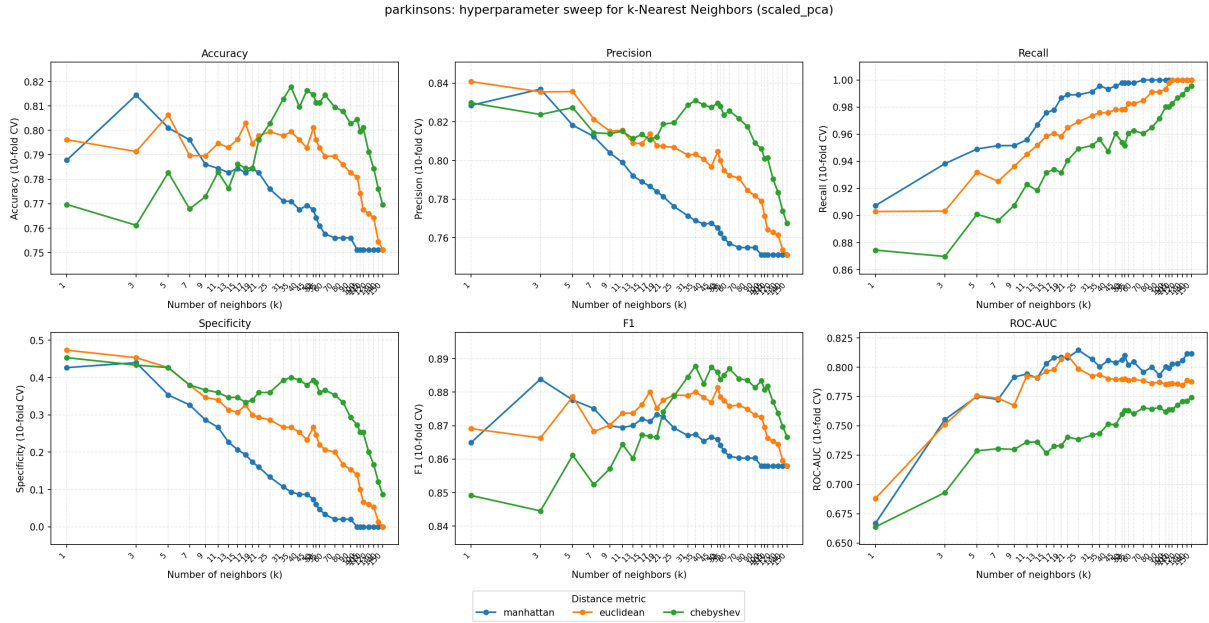


Figure 8: kNN hyperparameter sweep on Parkinson's: F1 and five other metrics versus the number of neighbors, segmented by distance metric.

Decision Tree. Figure 9 is a textbook overfitting curve. F1 and accuracy both peak at $\text{max_depth} = 2$ and drop monotonically as depth grows. The trade-off across metrics is striking: deeper trees lose F1 but *gain* precision and specificity — the deeper model becomes more selective, predicting PD less often. With only ~ 200 training samples and 753 features, the tree exhausts its splitting budget before $\text{depth} \approx 25$ and the curves flatten. Best F1: $\text{max_depth} = 2$, $\text{min_samples_leaf} = 0.001$, $F1 = 0.864$.

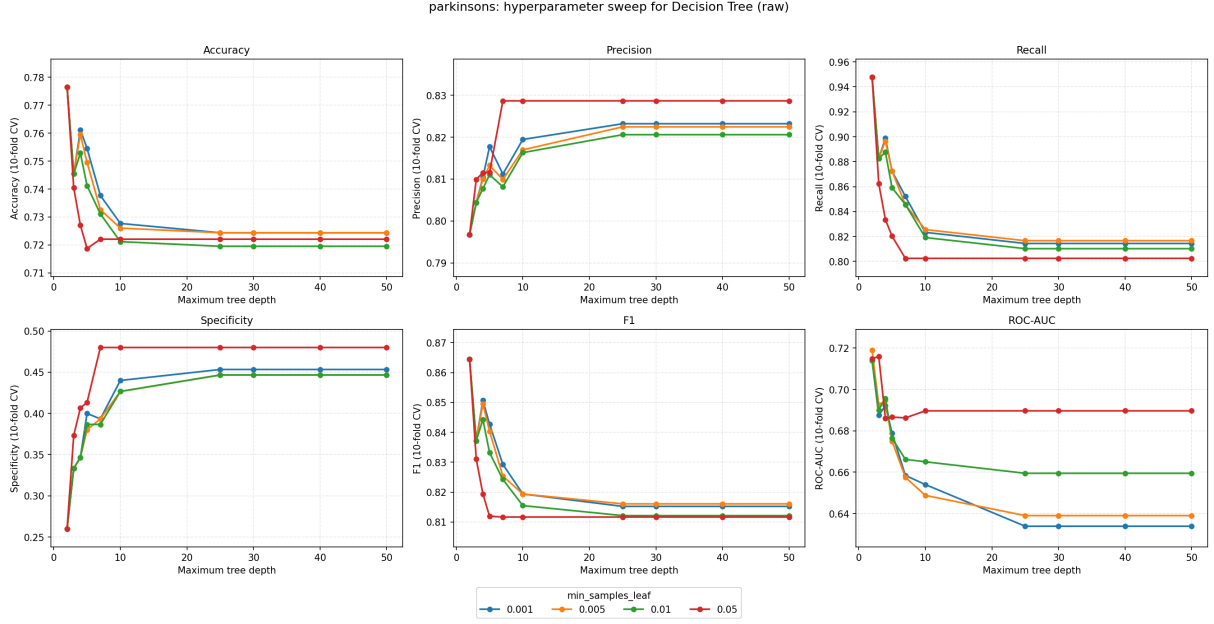


Figure 9: Decision Tree hyperparameter sweep on Parkinson's: ten `max_depth` values, four `min_samples_leaf` values, all six metrics.

Random Forest. Figure 10 shows F1 rising rapidly from 5 to about 100 trees and then saturating around $F1 \approx 0.89$. The `max_depth` lines for 25 and 70 are exactly superimposed — a direct observation that the individual trees stop splitting well before depth 25 (the dataset's pure-leaf criterion bites first), so any depth limit above ~ 25 is non-binding. ROC-AUC tracks F1 closely on this classifier, and specificity is the noisiest metric. Best F1: 100 trees, `max_depth` = 25, $F1 = 0.895$.

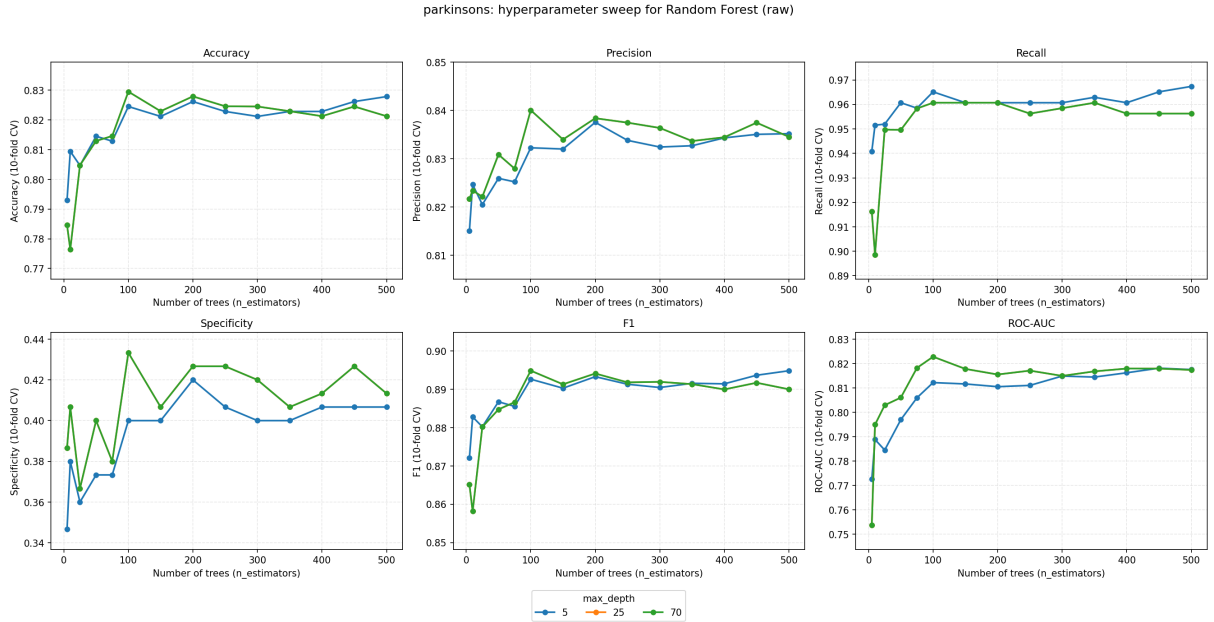


Figure 10: Random Forest hyperparameter sweep on Parkinson's: fourteen `n_estimators` values, three `max_depth` values.

Gradient Boosting. Figure 11 is the most informative panel for the multi-metric framing. F1 peaks at `n_estimators` = 50 with `max_depth` = 5 and then *declines slightly*, but ROC-AUC continues to climb past $n = 50$, peaking at $n = 250$. The `argmax` under F1 and under ROC-AUC

therefore disagree on this classifier — the F1-optimal model has ROC-AUC = 0.822, whereas the ROC-AUC-optimal model is more extensively boosted ($n = 250$) and trades a fraction of F1 for +0.017 ROC-AUC. Specificity climbs from 0.05 at $n = 5$ to ~ 0.47 at $n \geq 50$, the largest specificity range observed in any sweep. Shallow trees (`max_depth` = 5) dominate deeper ones across every metric. Best F1: $n = 50$, `max_depth` = 5, F1 = 0.900 — the highest F1 recorded on Parkinson’s.

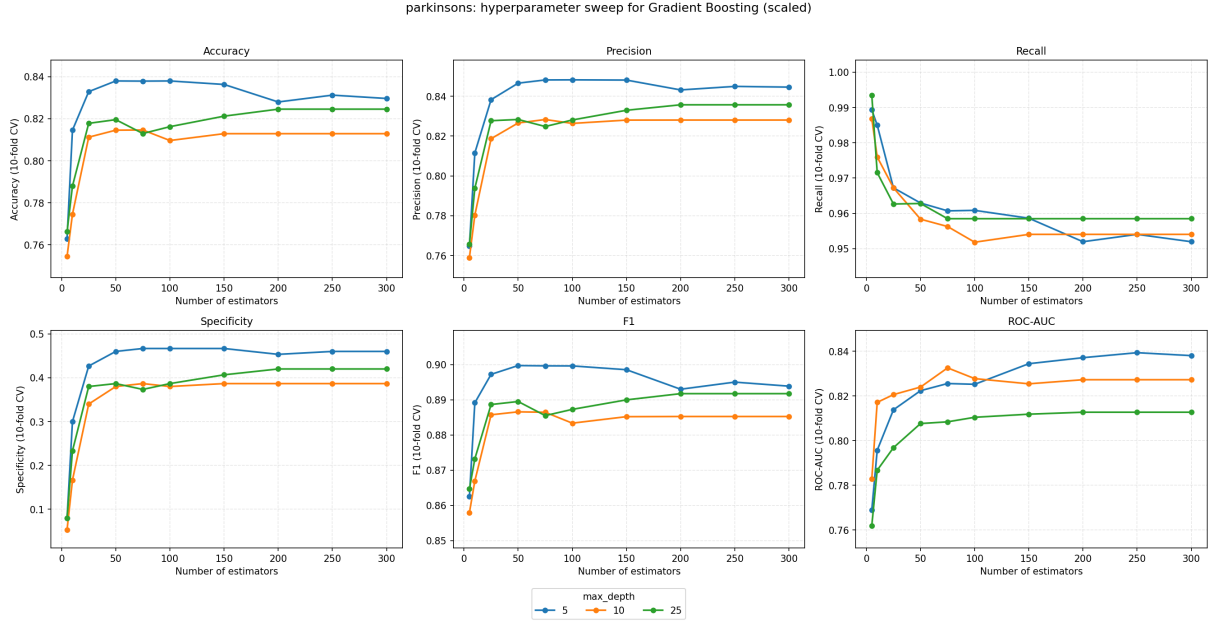


Figure 11: Gradient Boosting hyperparameter sweep on Parkinson’s: ten `n_estimators` values, three `max_depth` values.

XGBoost. Figure 12 has the smallest variation of any sweep (F1 spans only 0.873–0.891, a range of 0.018). The `max_depth` lines for 10, 25, and 50 are all coincident — the same depth-saturation effect that appears in Random Forest. Only `max_depth` = 5 traces a distinct curve, peaking at `n_estimators` = 75 with F1 = 0.891. The model is essentially insensitive to further tuning beyond the depth choice.

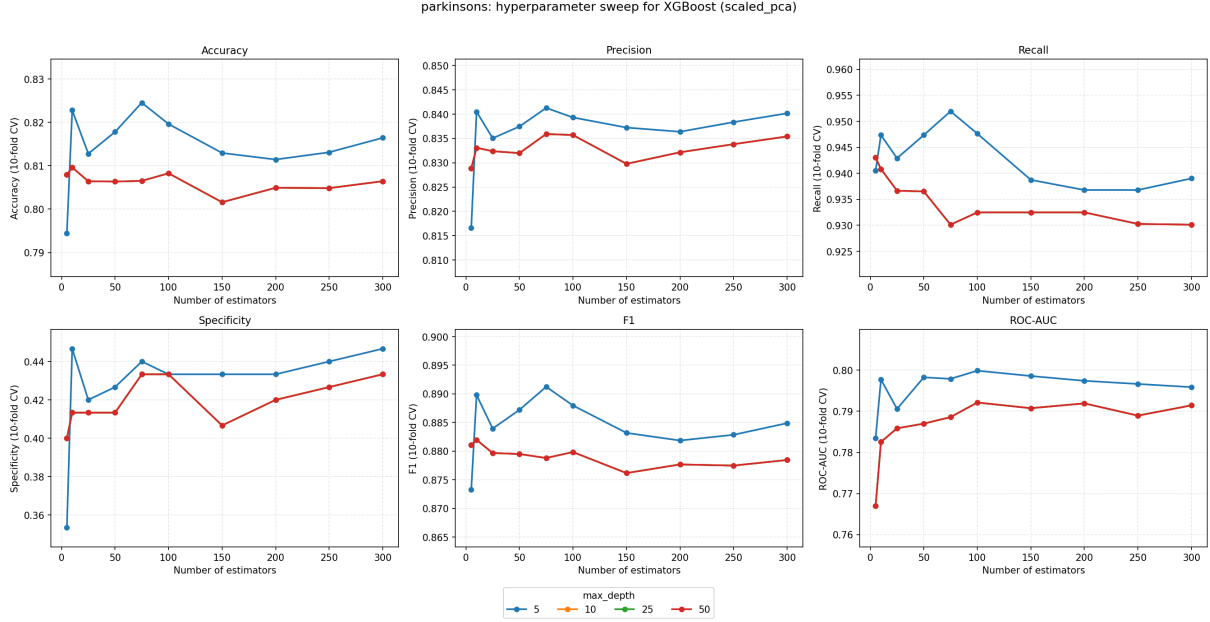


Figure 12: XGBoost hyperparameter sweep on Parkinson’s: ten `n_estimators` values, four `max_depth` values.

Cross-classifier synthesis. Ranking the five tunable classifiers by their best CV-F1: Gradient Boosting (0.900), Random Forest (0.895), XGBoost (0.891), kNN (0.886), Decision Tree (0.864). The four ensemble or instance-based methods cluster within 0.014 of each other — on this dataset, the choice of model family matters less than the choice of preprocessing (cf. Tables 4, 5, 7) and less than the choice of primary metric. Two recurring observations across the sweeps are worth flagging. First, the F1-optimal configuration is not always the ROC-AUC-optimal one (most visibly in Gradient Boosting and kNN): committing to a single primary metric quietly fixes a methodological choice. Second, the tree-based ensembles saturate their depth budget before reaching the larger `max_depth` limits on the sweep, so the deeper-line variants overlap exactly — a property of the dataset’s geometry (high feature count, low sample count) rather than of the algorithms themselves.

4.2.6 Final Parkinson’s Model

Stage 2 ran a per-classifier `GridSearchCV` on each model’s Stage-1-best preprocessing using the reduced grids in `configs/hyperparameters.yaml`. Table 9 summarises the tuned CV-F1 for each classifier. Gradient Boosting wins overall, edging Random Forest by 0.005 and XGBoost by 0.012. The full chosen configuration is Gradient Boosting on *scaled* features with `learning_rate` = 0.1, `max_depth` = 5, `n_estimators` = 50, `max_features` = $\sqrt{n}$.

Table 9: Per-classifier tuned CV-F1 from Stage 2 grid search on Parkinson’s. The chosen preprocessing is the Stage-1 winner from Table 3.

Classifier	Preprocessing	Tuned CV-F1	Rank
Naive Bayes	scaled + feature selection	0.8424	6
kNN	scaled + PCA	0.8859	4
Decision Tree	raw	0.8439	5
Random Forest	raw	0.8949	2
Gradient Boosting	scaled	0.8997	1
XGBoost	scaled + PCA	0.8880	3

Held-out test evaluation. Refitting the tuned Gradient Boosting on the full training set and evaluating on the 20% held-out test split ($n = 153$, of which 111 PD and 42 Healthy) produced the aggregate metrics in Table 10. The headline F1 of 0.905 is +0.051 above the majority-class baseline of 0.854 established in §4.1, and ROC-AUC of 0.938 indicates substantially better class separability than the F1 alone suggests — a recurring observation throughout this analysis.

Table 10: Test-set metrics for the tuned Gradient Boosting model on Parkinson’s. The *Healthy* and *PD* columns are the per-class decomposition; *Aggregate* is the binary (positive-class) view that F1, precision, and recall report by default.

Metric	Aggregate	Healthy	PD
Precision	0.833	0.952	0.833
Recall	0.991	0.476	0.991
F1	0.905	0.635	0.905
Support	153	42	111
Accuracy	0.850		
ROC-AUC	0.938		

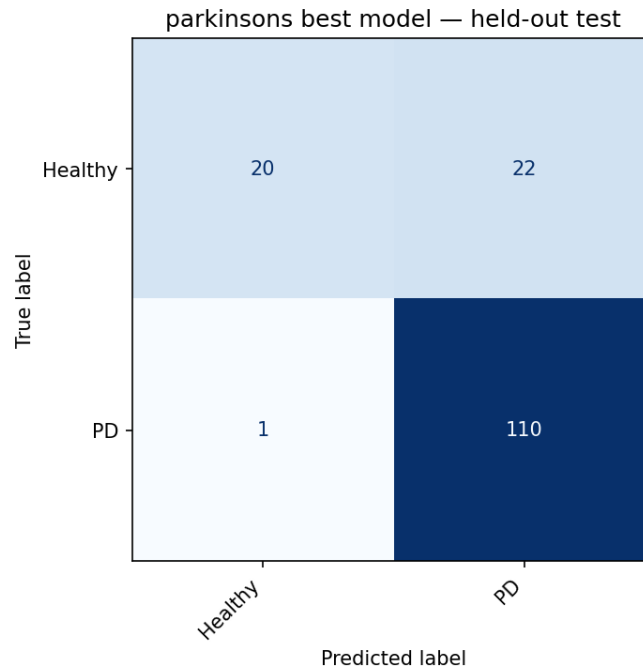


Figure 13: Confusion matrix for the tuned Gradient Boosting model on the held-out Parkinson’s test set. Rows are true labels, columns are predictions.

The confusion matrix in Figure 13 exposes the per-class asymmetry that the aggregate F1 hides. Of 111 PD test patients, the model correctly identifies 110 (recall = 0.991); only one PD case is missed. Of 42 Healthy test patients, however, the model correctly identifies only 20 (specificity = 0.476): more than half of the healthy patients are predicted as PD. Per-class F1 reflects this: PD F1 = 0.905 versus Healthy F1 = 0.635.

This pattern is not unique to the chosen model. Section 4.2.5 showed recall sitting at 0.95–1.00 across nearly every cell of every classifier’s hyperparameter sweep while specificity swung between 0 and 0.47. The 75/25 class distribution biases all six classifier families toward predicting PD aggressively, and threshold tuning rather than further hyperparameter search would be the next lever for trading PD recall against Healthy specificity. The ROC-AUC of 0.938 confirms that

this trade-off exists in the model’s score distribution: there is real separability available; the default decision threshold simply sits closer to the high-recall end of the operating curve than the symmetric point.

4.3 Covertypes Results

This section mirrors the Parkinson’s results, but with Covertypes-specific metrics. Since Covertypes is a multiclass classification problem, accuracy is reported together with macro-averaged metrics and per-class results.

4.3.1 Overall Model Comparison

Table 11: Overall comparison of the six classifiers on Covertypes, each evaluated with its Stage-2 best preprocessing and tuned hyperparameters. Metrics are 10-fold cross-validation means on the training set.

Model	Best preprocessing	Accuracy	Macro F1	Macro Precision	Macro Recall
Naive Bayes	raw	0.5996	0.5528	0.6526	0.5996
kNN	raw	0.7943	0.7885	0.7885	0.7943
Decision Tree	raw	0.7380	0.7365	0.7374	0.7380
Random Forest	scaled + FS	0.8271	0.8240	0.8248	0.8271
Gradient Boosting	raw	0.8304	0.8285	0.8290	0.8304
XGBoost	raw	0.8305	0.8281	0.8287	0.8305

The three tree-based ensembles cluster within 0.003 macro F1 of each other (Random Forest 0.824, XGBoost 0.828, Gradient Boosting 0.829): on this dataset the choice among them is essentially a coin flip, and the selection of Gradient Boosting in §4.3.5 reflects a CV margin that is much smaller than a single standard deviation. kNN at $k = 1$ follows at macro F1 = 0.789, Decision Tree at 0.737, and Gaussian Naive Bayes at 0.553 — well above the balanced-class random baseline of $1/7 \approx 0.143$ but the only model that fails to cross macro F1 = 0.7. A signature of the balanced subsampling carried out in §2.2 is visible in the table: accuracy equals macro recall to four decimal places for every classifier (because micro and macro recall coincide when every class has equal support), so on this dataset the two columns carry the same information.

4.3.2 Effect of Scaling

Table 12: Effect of scaling on Covertypes: cross-validation mean macro F1 for each classifier under raw and scaled features.

Model	Raw Macro F1	Scaled Macro F1	Difference
Naive Bayes	0.5528	0.3748	−0.1779
kNN	0.7396	0.7280	−0.0117
Decision Tree	0.7344	0.7339	−0.0005
Random Forest	0.8173	0.8175	+0.0003
Gradient Boosting	0.7750	0.7749	−0.0002
XGBoost	0.8247	0.8247	+0.0000

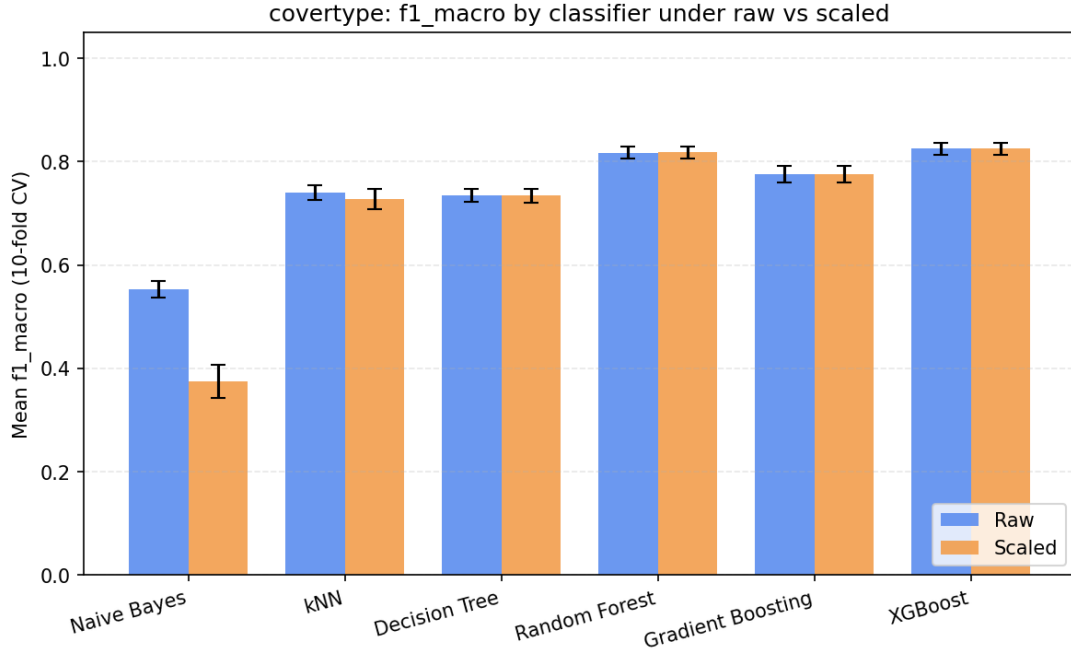


Figure 14: Macro F1-score for each classifier on Covertime under raw versus scaled preprocessing.

Scaling produces the largest single negative effect observed in either dataset’s preprocessing analysis: Naive Bayes loses nearly eighteen macro F1 points ($0.553 \rightarrow 0.375$) under standard scaling. This is consistent with the Covertime mixed feature structure — ten continuous physical measurements (elevation, slope, distances) and forty-four one-hot binary indicators (wilderness area, soil type). Rescaling the continuous columns to mean zero, unit variance creates a representational mismatch with the unrescaled 0/1 binary columns; Gaussian Naive Bayes’s per-feature Gaussian assumption then no longer applies coherently across the feature space, distorting the posterior computation.

The remaining five classifiers are essentially unaffected by scaling (differences within ± 0.012 macro F1), and four of them register zero change to four decimal places. For Covertime, the conclusion of the scaling experiment is therefore simple: scaling provides no benefit to any classifier and substantially damages Naive Bayes, so the raw-features configuration is the safer default. The mechanism underlying the kNN result specifically — why a preprocessing step that yielded a large gain on Parkinson’s produces no effect here — is examined in the cross-dataset discussion of §4.4, where the feature-level breakdowns in §4.2.3 and §4.3.3 are available.

4.3.3 Feature Selection Analysis

Table 13: Feature selection at $k = 30$ (of 54 available features) versus the scaled baseline.

Model	Scaled Macro F1	Scaled + FS Macro F1	Difference
Naive Bayes	0.3748	0.3944	+0.0195
kNN	0.7280	0.7339	+0.0060
Decision Tree	0.7339	0.7327	−0.0012
Random Forest	0.8175	0.8202	+0.0027
Gradient Boosting	0.7749	0.7720	−0.0029
XGBoost	0.8247	0.8208	−0.0039

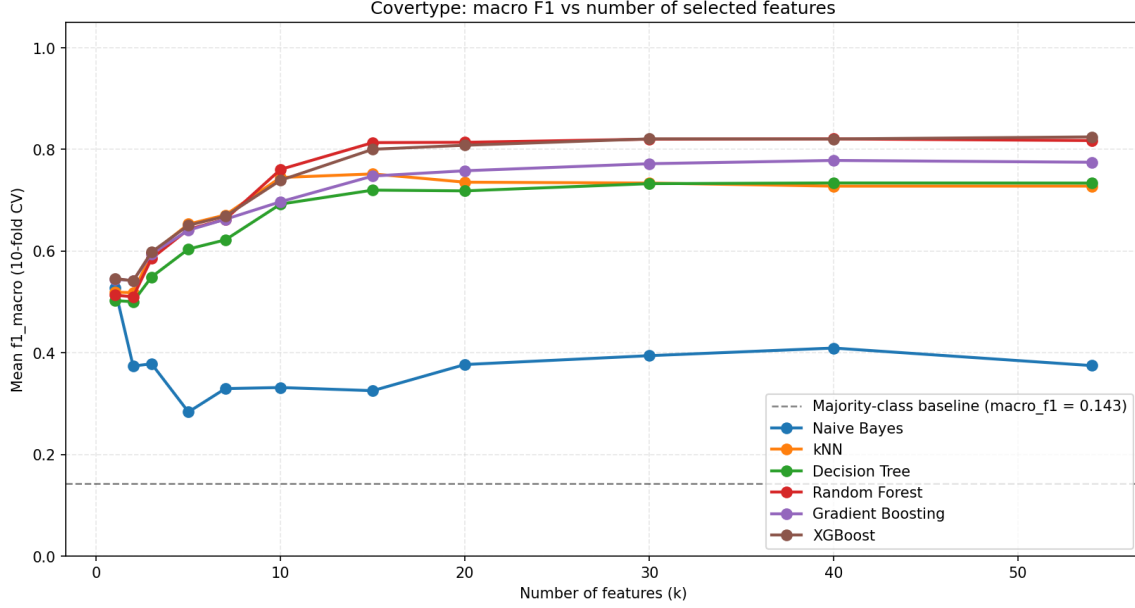


Figure 15: Macro F1 versus number of selected features for all six classifiers on Covtype.

Feature selection with $k = 30$ produces small or negligible effects for every classifier. The largest positive effect — for Naive Bayes — is a partial recovery from the scaling damage rather than a true improvement: the absolute macro F1 of 0.394 is still far below the raw-feature baseline of 0.553. For the tree-based ensembles, differences are within ± 0.004 macro F1 — well within fold-to-fold noise.

The feature-selection sweep across $k \in \{1, \dots, 54\}$ (Figure 15) shows that all ensembles reach near-peak performance by $k = 15$ and remain flat thereafter, indicating that fewer than a third of the 54 available features carry the bulk of the discriminative signal.

Which features drive the signal? To identify those features explicitly, the ANOVA F-statistic was computed on the (scaled) training set for each of the 54 columns. Table 14 lists the top 15 by F-score, together with the family each feature belongs to. Two findings dominate the table.

First, **Elevation** alone has an F-statistic of 6113.8, which is $3.3\times$ the second-ranked feature (**Wilderness_Area4**, $F = 1852.2$) and more than $12\times$ the third-ranked feature (**Horizontal_Distance_To_Roadway**, $F = 487.3$). No other column comes close to Elevation’s univariate discriminative power on this dataset.

Second, the top 15 features are split across all three feature families present in the dataset: 5 continuous cartographic measurements (including all four distance-based measurements that rank in the top 15), 3 of the 4 wilderness-area binary indicators, and 7 of the 40 soil-type binary indicators. So the contention is not that “useful features are concentrated in the continuous columns”: both binary families contribute strongly. Each family does, however, contribute at a different scale — 5/10 continuous features land in the top 15 (50% of the family), against 3/4 of the wilderness areas (75%) and only 7/40 of the soil types (17.5%). The soil-type family is the most heterogeneous: a handful are highly discriminative, but the majority rank in the bottom half by F-statistic.

The F-statistic falls off rapidly past the top 15: by rank 20 it is at roughly 1% of rank 1’s value, and the remaining ~ 35 features together account for very little incremental signal. This is exactly the $k = 15$ saturation point visible in Figure 15.

Table 14: The 15 most discriminative individual features on the Covertypes training set, ranked by the ANOVA F-statistic (computed on the scaled training data of the balanced subsample, 80/20 stratified split). Family membership uses the partitioning described in §2.2.

Rank	Feature	Family	F-score
1	Elevation	Continuous cartographic	6113.8
2	Wilderness_Area4	Wilderness area	1852.2
3	Horizontal_Distance_To_Roadways	Continuous cartographic	487.3
4	Soil_Type3	Soil type	371.0
5	Soil_Type10	Soil type	349.7
6	Wilderness_Area1	Wilderness area	324.1
7	Soil_Type38	Soil type	301.9
8	Horizontal_Distance_To_Fire_Points	Continuous cartographic	286.4
9	Soil_Type39	Soil type	250.9
10	Wilderness_Area3	Wilderness area	171.0
11	Hillshade_9am	Continuous cartographic	168.6
12	Horizontal_Distance_To_Hydrology	Continuous cartographic	146.6
13	Soil_Type30	Soil type	142.0
14	Soil_Type40	Soil type	139.9
15	Soil_Type29	Soil type	122.2

Principal Component Analysis. PCA was not evaluated for Covertypes. As justified in Section 3.5.2, the dataset’s mixed continuous and one-hot encoded structure makes PCA components dominated by the variance of the continuous columns; the resulting axes would not correspond to interpretable cartographic categories.

4.3.4 Hyperparameter Sweep Analysis

For each tunable classifier (kNN, Decision Tree, Random Forest, Gradient Boosting, and XGBoost) the two principal hyperparameters were swept using 10-fold stratified CV on the balanced training subsample, with the secondary hyperparameter shown as separate lines on each plot and the remaining hyperparameters held at their defaults. Each classifier was evaluated on its Stage-2 best preprocessing configuration. Naive Bayes is omitted: the variant used has no tunable hyperparameters in the search space considered. All four scoring metrics — accuracy, macro precision, macro recall, and macro F1 — are recorded for every cell.

k-Nearest Neighbors. Figure 16 sweeps k across 30 values from 1 to 150. Every metric decreases monotonically with k for all three distance metrics, with Manhattan above Euclidean above Chebyshev throughout. The best configuration sits at the smallest possible k : $k = 1$, Manhattan, macro F1 = 0.789. The range across the sweep is unusually large for a single classifier — 0.488 to 0.789, a swing of 0.30 macro F1. With $\sim 5,600$ balanced training samples and seven well-separated cover types, smoothing across neighbors pulls in samples from the wrong class faster than it filters noise; the single nearest neighbor is the best predictor available.

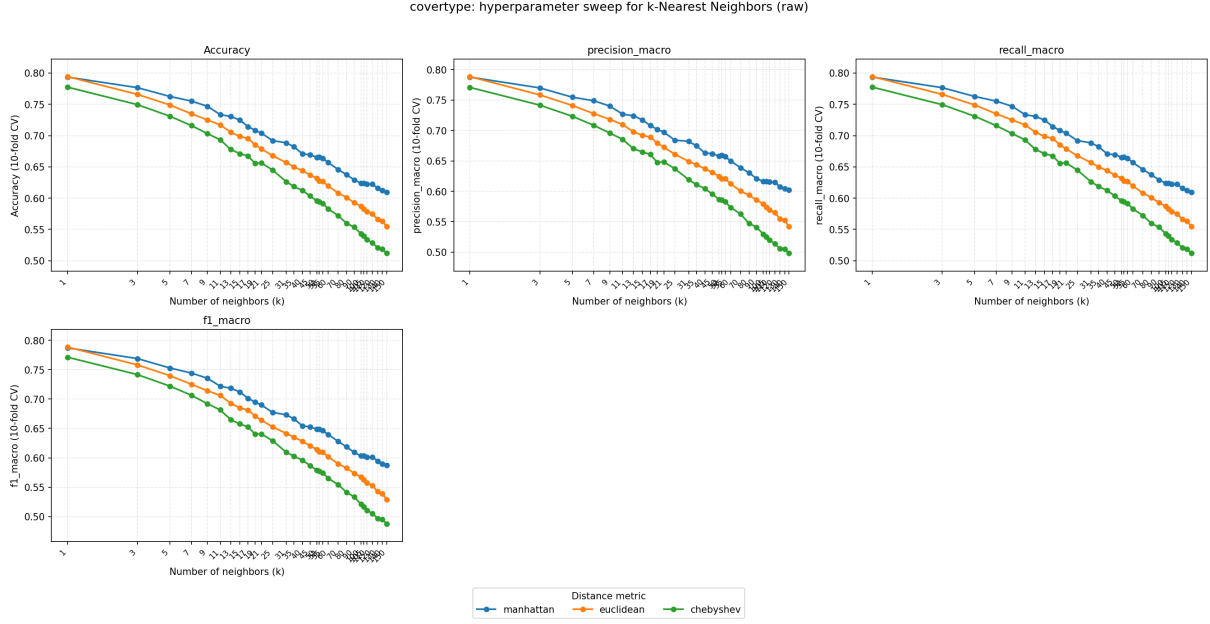


Figure 16: kNN hyperparameter sweep on Covertime: macro F1 and the three other scoring metrics versus the number of neighbors, segmented by distance metric.

Decision Tree. Figure 17 shows the largest range of any classifier on either dataset: macro F1 spans 0.344 at `max_depth` = 2 to 0.719 at `max_depth` = 10. Three of the four `min_samples_leaf` settings (0.001, 0.005, 0.01) converge once depth exceeds ~ 10 , while `min_samples_leaf` = 0.05 (requiring at least ~ 280 samples per leaf) plateaus much earlier and much lower, around F1 = 0.59. The best configuration is `max_depth` = 10, `min_samples_leaf` = 0.001, F1 = 0.719. The contrast with the corresponding Parkinson's sweep — where `max_depth` = 2 won — comes from the sample-count asymmetry: deeper trees overfit Parkinson's ~ 200 training samples but find more usable splits on Covertime's $\sim 5,600$.

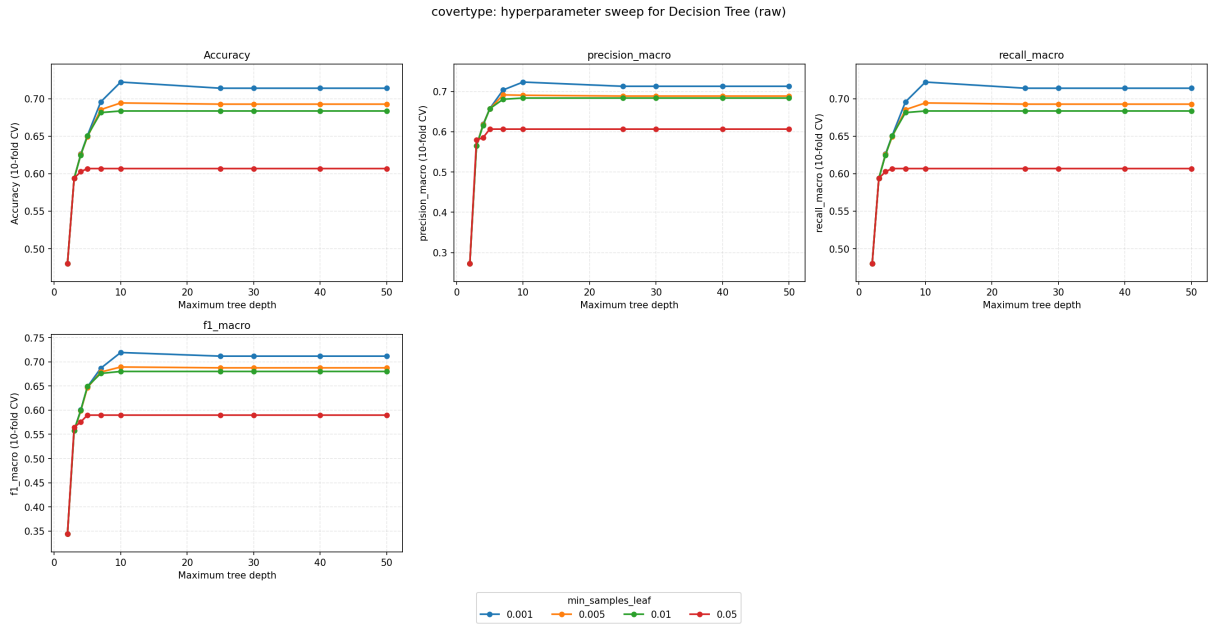


Figure 17: Decision Tree hyperparameter sweep on Covertime: ten `max_depth` values, four `min_samples_leaf` values.

Random Forest. Figure 18 shows two distinct regimes. The `max_depth = 5` line plateaus around macro F1 = 0.70 regardless of `n_estimators`, confirming that depth 5 is too shallow to fit the Covertypes decision boundary. The `max_depth = 25` and `max_depth = 70` lines, in contrast, are exactly coincident across every `n_estimators` value — the same depth-saturation effect as in the Parkinson’s RF sweep, where individual trees stop splitting because of pure leaves long before reaching the depth ceiling. F1 rises from ~ 0.76 at 10 trees to ~ 0.82 at 100 and gains little thereafter. Best configuration: 450 trees, `max_depth = 70`, F1 = 0.828.

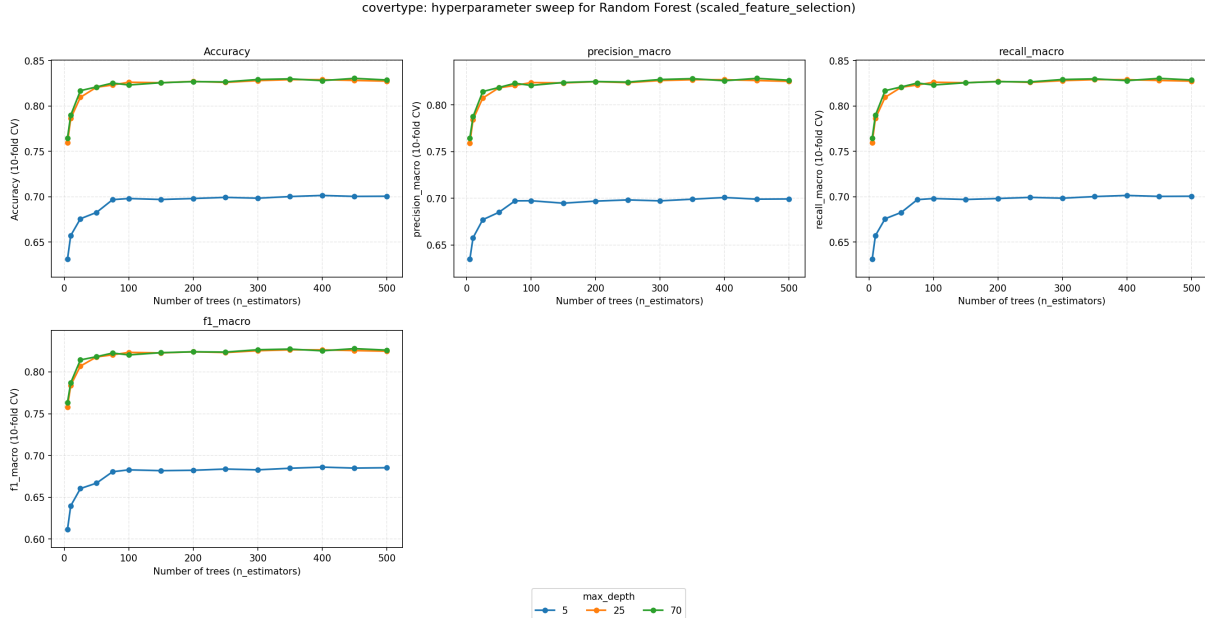


Figure 18: Random Forest hyperparameter sweep on Covertypes: fourteen `n_estimators` values, three `max_depth` values.

Gradient Boosting. Figure 19 shows that `max_depth = 10` dominates both shallower (depth 5) and deeper (depth 25) alternatives across the full range of `n_estimators`. Macro F1 rises monotonically with `n_estimators` for depth 10, reaching 0.832 at 300 estimators and showing no plateau within the swept range — a higher count might give further (small) gains. Depth 25 is competitive at low `n_estimators` but is overtaken by depth 10 from about 100 estimators onward and shows clear signs of depth saturation past $n = 150$ (F1 frozen at 0.825). Gradient Boosting attains the highest macro F1 of any classifier on Covertypes, narrowly: 0.832 versus XGBoost’s 0.831 and Random Forest’s 0.828.

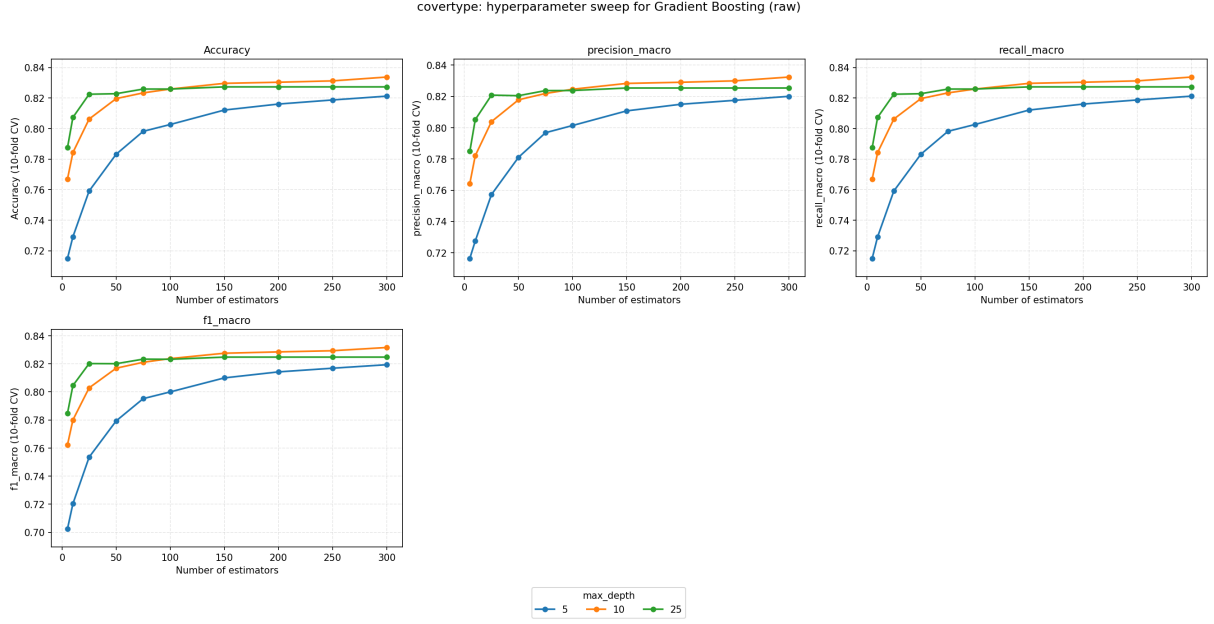


Figure 19: Gradient Boosting hyperparameter sweep on Covertypes: ten `n_estimators` values, three `max_depth` values.

XGBoost. Figure 20 shows the smallest variation of any classifier on Covertypes (macro F1 range 0.107) but still reaches near the top of the overall ranking. Depth 5 under-fits and plateaus at $F1 \approx 0.81$; depths 10, 25, and 50 rise together rapidly and become essentially indistinguishable past $n = 100$, with depths 25 and 50 exactly coincident throughout (the same saturation effect as in Random Forest). The best cell is `n_estimators` = 100, `max_depth` = 25, $F1 = 0.831$ — effectively tied with Gradient Boosting.

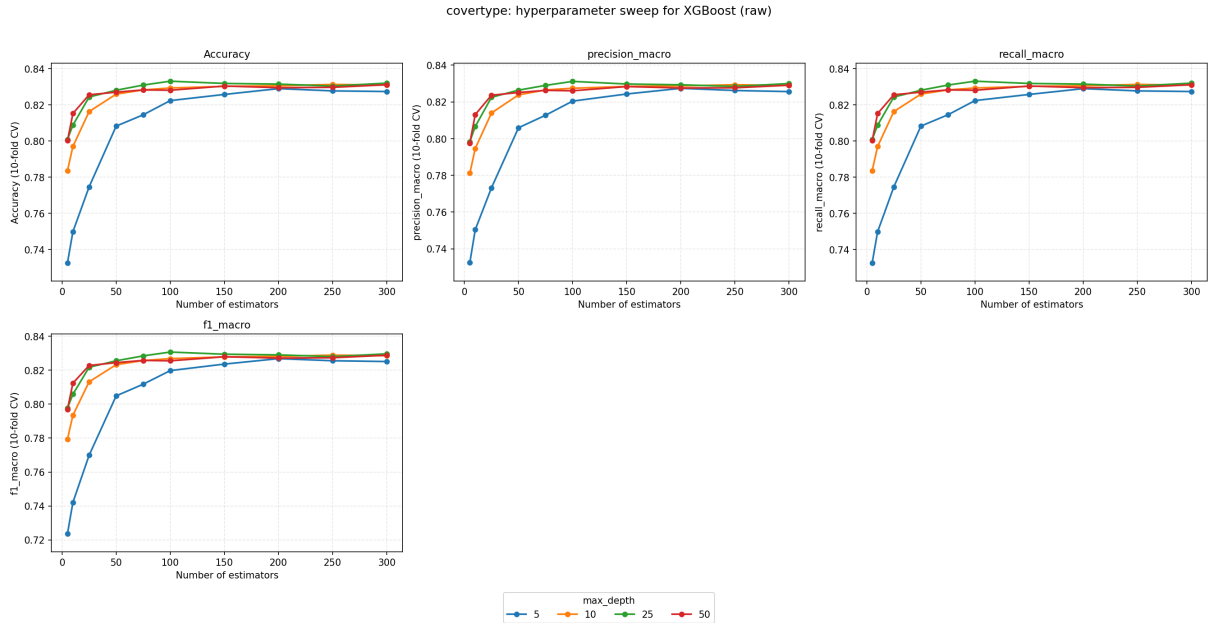


Figure 20: XGBoost hyperparameter sweep on Covertypes: ten `n_estimators` values, four `max_depth` values.

Cross-classifier synthesis. Ranking the five tunable classifiers by their best CV macro F1 across these sweeps: Gradient Boosting (0.832), XGBoost (0.831), Random Forest (0.828), kNN (0.789), Decision Tree (0.719). The three tree-based ensembles cluster within 0.005 of each other

— essentially indistinguishable on this dataset — and beat the single Decision Tree by about 0.11 F1, the largest model-family gap of any sweep in this study. Two further observations are worth flagging. First, hyperparameter sensitivity on Covertypes is far higher than on Parkinson’s: even the most-stable classifier (XGBoost) spans 0.107 F1 across the swept grid, against Parkinson’s XGBoost which spanned only 0.018. Most of this is driven by the under-fit end of the curves — shallow trees (5), tiny ensembles ($n \leq 10$), or over-restrictive `min_samples_leaf` — but the magnitude shows that on a larger and more multi-class dataset, the choice of hyperparameter is a much more consequential decision than the choice of preprocessing configuration (cf. Table 11). Second, the same depth-saturation pattern observed on Parkinson’s — where individual trees stop splitting before reaching the depth ceiling — recurs on Covertypes’s Random Forest and XGBoost; this appears to be a property of the `ColumnTransformer`-based training data rather than the algorithm, and any `max_depth` value above ~ 25 should be treated as “no depth limit”.

4.3.5 Final Covertypes Model and Per-Class Performance

The best-performing model on Covertypes was Gradient Boosting on raw features, with tuned hyperparameters `learning_rate` = 0.1, `max_depth` = 10, `n_estimators` = 200, and `max_features` = $\sqrt{n}$. Refitting this configuration on the full training set and evaluating on the held-out 20% test split ($n = 1,400$, 200 per class) produced the aggregate metrics in Table 15.

Table 15: Test-set metrics for the tuned Gradient Boosting model on Covertypes, broken down per class. The bottom block shows overall accuracy and the macro-averaged metrics. Classes are listed in their original label order (1–7).

Class	Precision	Recall	F1	Support
Spruce/Fir	0.768	0.695	0.730	200
Lodgepole Pine	0.680	0.595	0.635	200
Ponderosa Pine	0.857	0.780	0.817	200
Cottonwood/Willow	0.925	0.985	0.954	200
Aspen	0.786	0.935	0.854	200
Douglas-fir	0.826	0.855	0.840	200
Krummholz	0.926	0.945	0.936	200
Accuracy			0.827	1,400
Macro average	0.824	0.827	0.824	1,400

The macro F1 of 0.824 is well above the seven-class balanced-class random baseline of $1/7 \approx 0.143$ and tracks the cross-validation estimate of 0.829 from Table 11 within 0.005, a normal held-out drop. Accuracy and macro recall coincide for the same reason noted in the discussion below Table 11: every class has equal support on the held-out split, so micro and macro recall coincide.

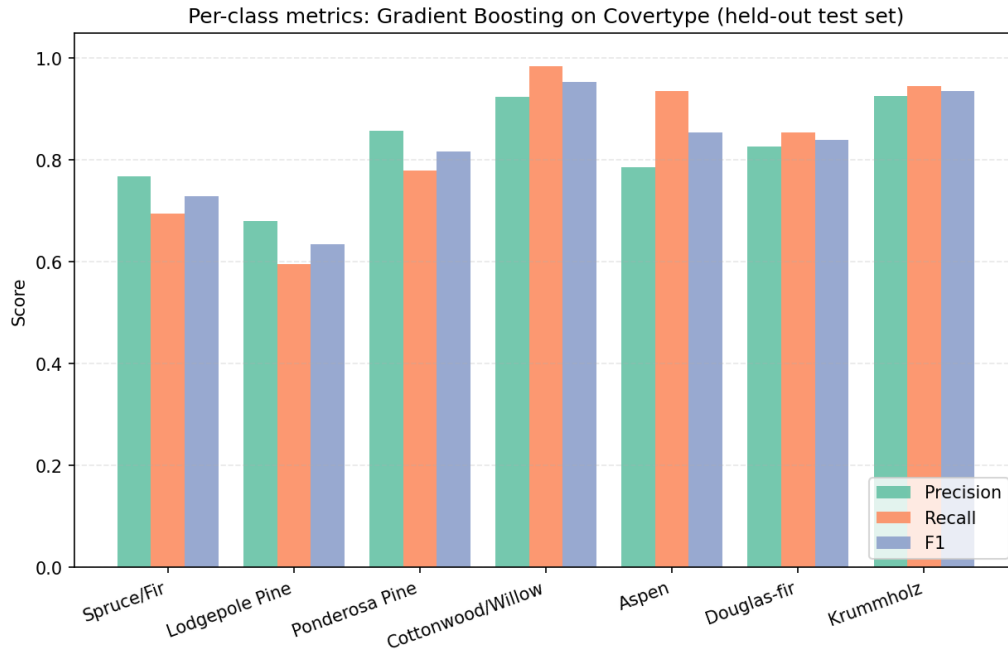


Figure 21: Per-class precision, recall, and F1-score for the best Covertypes model on the held-out test set. Cottonwood/Willow and Krummholz achieve F1 above 0.94, while Lodgepole Pine plateaus around 0.63 due to systematic confusion with Spruce/Fir.

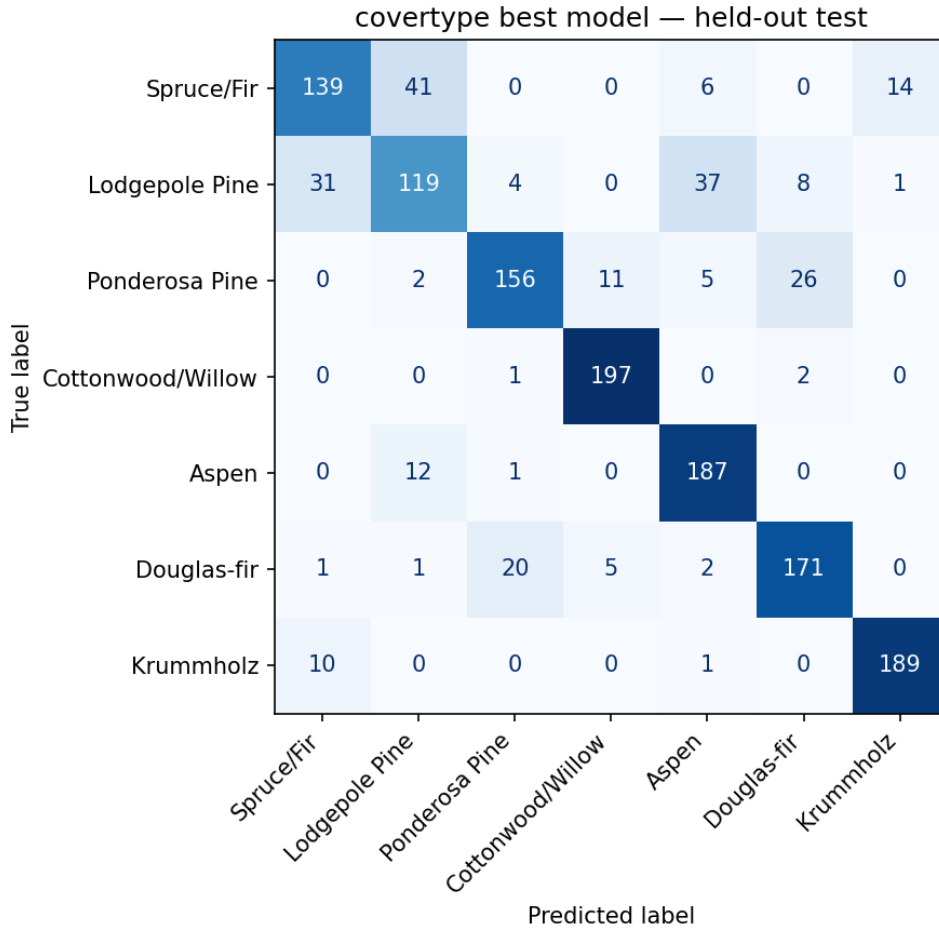


Figure 22: Confusion matrix for the best Covertypes model on the held-out test set. The systematic confusion between Spruce/Fir and Lodgepole Pine is visible as the off-diagonal entries in the top-left of the matrix.

The per-class results in Figure 21 reveal three distinct behavioural tiers among the seven cover types:

- **Cottonwood/Willow** (F1 0.95) and **Krummholz** (F1 0.94) classify cleanly. Cottonwood/Willow grows near streams, captured by the distance-to-hydrology feature; Krummholz is the high-elevation tree-line species, captured by elevation alone.
- **Aspen**, **Douglas-fir**, **Ponderosa Pine**, **Spruce/Fir** occupy a middle band with F1 between 0.73 and 0.85.
- **Lodgepole Pine** (F1 0.63) is the worst-performing class. This is the dominant source of error in the confusion matrix: 41 actual Spruce/Fir samples are misclassified as Lodgepole Pine, and 31 actual Lodgepole Pine samples are misclassified as Spruce/Fir. The two species coexist at similar elevations with overlapping cartographic profiles, so the available features do not separate them cleanly.

The macro F1 ceiling of 0.82 is bounded not by classifier capacity but by intrinsic feature limitations: no amount of feature engineering can separate Lodgepole Pine from Spruce/Fir using the cartographic variables available. Improving this would require new features (e.g., tree morphology, soil moisture, climate normals) rather than further model tuning.

4.4 Comparison Between Datasets

Gradient Boosting wins on both datasets: $F1 = 0.900$ on Parkinson’s (Table 3) and macro $F1 = 0.829$ on Covertypes (Table 11). On both, the tree-based ensembles — Random Forest, Gradient Boosting, XGBoost — cluster tightly at the top (≤ 0.005 apart within each dataset). The gap between the ensembles and the simpler classifiers (Decision Tree, Naive Bayes), however, is much wider on Covertypes than on Parkinson’s: Covertypes’s $\sim 5,600$ training samples give the ensembles more room to exploit the bias–variance trade-off than Parkinson’s ~ 200 do.

Preprocessing has a measurably larger effect on Parkinson’s, while hyperparameter choice has a much larger effect on Covertypes. The largest single preprocessing effect in either dataset is the $+0.06$ $F1$ gain that scaling produces for kNN on Parkinson’s (against essentially zero on Covertypes); the largest single hyperparameter sweep range is the 0.37 Decision Tree macro $F1$ spread on Covertypes (against 0.05 on Parkinson’s). The same intervention can also flip in optimal value: Decision Tree’s best `max_depth` is 2 on Parkinson’s (where deeper trees overfit) and 10 on Covertypes (where shallower trees underfit), and kNN’s best k is 35–50 on Parkinson’s (smoothing across neighbours helps) and 1 on Covertypes (smoothing pulls in samples from the wrong class). Sample count is the unifying explanation.

The two datasets impose qualitatively different evaluation challenges. Parkinson’s 75/25 class imbalance biases every classifier toward predicting PD: recall sits above 0.95 for the top four classifiers, while Healthy-class specificity is much lower (the chosen model reaches PD recall 0.99 but Healthy recall only 0.48). Covertypes’s seven balanced classes avoid this aggregate asymmetry but reveal a feature-limited per-class ceiling: Lodgepole Pine and Spruce/Fir share enough of their cartographic profile that the final model confuses them systematically (Figure 22), capping macro $F1$ around 0.82 regardless of hyperparameter choice. On Parkinson’s, the next lever would be threshold tuning or class weighting; on Covertypes, additional features.

5 Conclusion

5.1 Summary of Main Findings

This study compared preprocessing strategies and classification algorithms on two different classification problems: Parkinson’s disease prediction from speech recordings and forest cover type prediction from cartographic variables. The results showed that model performance depends strongly on both the dataset characteristics and the classifier family. Tree-based ensemble methods, especially Gradient Boosting, Random Forest, and XGBoost, were the most consistent performers across both datasets.

5.2 Parkinson’s Disease Findings

For the Parkinson’s disease dataset, Gradient Boosting achieved the strongest tuned performance and was selected as the final model. The model achieved high overall performance and very high recall for Parkinson’s disease cases, meaning that it correctly identified most PD recordings. However, the per-class analysis showed that healthy controls were much harder to classify. The model tended to favour the PD class, which is consistent with the moderate class imbalance in the dataset.

The preprocessing analysis showed that scaling was especially important for k-Nearest Neighbors, while tree-based models were mostly unaffected by scaling. Feature selection and PCA confirmed that the Parkinson’s feature space contains substantial redundancy, since competitive performance could be maintained after reducing the number of features. However, these dimensionality-reduction methods did not fully solve the weaker healthy-control detection.

5.3 Covertypes Findings

For the Covertypes dataset, Gradient Boosting, XGBoost, and Random Forest achieved the strongest results on the balanced 7,000-record subset. The final model clearly outperformed the majority-class baseline and achieved strong macro-level performance. Scaling had little benefit for most models and even harmed some classifiers, showing that preprocessing effects do not transfer uniformly across datasets.

The per-class results showed that some cover types were easier to classify than others. Cottonwood/Willow and Krummholz achieved the strongest class-level performance, while Lodgepole Pine and Spruce/Fir were more frequently confused. This suggests that the available cartographic features do not separate all forest classes equally well.

5.4 Limitations

The main limitation of the Parkinson’s experiment is that the final model achieved much stronger performance on the PD class than on the healthy-control class. Therefore, high overall F1-score should not be interpreted as balanced performance across both classes. For the Covertypes dataset, the main limitation is that the experiments were conducted on a balanced undersampled subset rather than the original imbalanced dataset. As a result, the reported Covertypes performance reflects the controlled balanced setting, not necessarily performance on the natural class distribution.

5.5 Future Work

Future work on the Parkinson’s dataset should investigate imbalance-aware strategies such as threshold tuning or class weighting. These approaches may improve healthy-control recall while preserving strong PD detection. For the Covertypes dataset, future work should evaluate the selected models on the original imbalanced distribution and investigate whether additional geographic or ecological variables could reduce confusion between similar classes such as Lodgepole Pine and Spruce/Fir.